

Simulated Upwork Order — OTC Pharmacy Multi-Branch Sales Analysis

(Practice brief — treat this exactly like a real hired job. Everything below is what a client would realistically send you after hiring.)

Job Post (as it would appear on Upwork)

Title: OTC Pharmacy Chain — Sales EDA & Power BI Dashboard (7 Branches, FY2022)

Category: Data Analysis, Business Intelligence, Power BI

Budget: Fixed price — \$450-\$650 (client open to hourly for revisions)

Client: Meridian Health Retail Group — a mid-size OTC pharmacy retail chain operating 7 branches. Client posts as "Operations & Insights Lead," not a data person themselves.

Job description (as posted):

We run 7 pharmacy branches and have never had a proper look at our 2022 sales beyond basic spreadsheet totals our regional managers send monthly. We recently combined our raw sales export with some branch-level demographic data (population, median income, nearby healthcare centers) but nobody on our team has time to actually dig into it.

We need someone who can (1) clean and explore the data properly, and (2) build us a Power BI dashboard we can actually use in our quarterly ops review — not just pretty charts, but something that helps us decide where to stock more dermatology/topical products, which branches are underperforming relative to their local market, and whether our reps are being used efficiently.

Please include a short written summary of your key findings along with the dashboard. Bonus if you show your data cleaning process (Power Query or Python) since we want to understand how you got from raw data to final numbers, not just receive a black box.

Client's First Message After Hiring You

Hi! Excited to get started. A bit more context on what we're hoping to see, so you're not guessing:

What's been bugging us:

1. We think our topical/dermatology products (creams etc.) sell disproportionately at our Lakeview, Gold Coast, and Wicker Park locations, but we've never actually confirmed that with numbers.

2. We don't know if our higher-population branches (like Austin) are actually underperforming *relative to their size* — raw revenue numbers make them look fine, but we suspect they're leaving money on the table.
3. Is there any real relationship between a branch's local median income and what it sells? We want to know if we should be tailoring branch inventory by neighborhood income level.
4. We want to know if any of our sales reps are carrying certain branches or categories way more than others — useful for our performance reviews next quarter.
5. How does revenue move month-to-month? Is there seasonality we should be stocking ahead of (e.g., before flu season, before summer)?
6. We switched some branches to accept more insurance-based payments this year — did that change anything about basket size or category mix?
7. Which single category is our best revenue driver, and which one is basically dead weight?

What we want delivered:

- A short written EDA summary (1-2 pages max, we don't want a thesis — just tell us what you found and why it matters)
- A Power BI dashboard we can filter by branch, category, date range, and payment method
- Please don't just show us totals — we want per-capita and per-healthcare-center context since our branches are very different sizes
- If you spot any data quality issues (weird outliers, missing values, anything that looks synthetic/off), flag them — we'd rather know than have a dashboard that hides problems

✓ **What This Client Is Really Asking For (translated into deliverables)**

Client's question	What you'll need to build
Are BR-05/06/07 really driving cream sales?	Category mix by branch (stacked bar or 100% stacked bar), statistical check (e.g., chi-square or simple % comparison)
Is Austin underperforming for its size?	Revenue per capita and revenue per healthcare center metrics — not just raw revenue
Does income correlate with spend?	Scatter plot: median income vs. revenue/branch, plus a correlation coefficient

Client's question	What you'll need to build
Are reps unevenly loaded?	Sales rep performance matrix — revenue by rep $\times$ branch, maybe a heatmap
Monthly seasonality?	Time series line chart, month-over-month % change, maybe rolling average
Payment method impact on basket size?	Avg order value (Amount $\div$ Boxes, or per transaction) split by Payment Method
Best/worst category?	Ranked bar chart of revenue by category, with box-shipped volume as a secondary metric
Data quality flags	A short "data caveats" section in your EDA — outliers, any synthetic-looking patterns, missing values

Suggested Project Structure (for your practice workflow)

1. Power Query / Python cleaning layer

- Document type conversions, merge logic (sales $\bowtie$ branch demographics on Branch ID), any deduplication or outlier handling
- This is what the client explicitly asked to see — don't skip documenting it

2. EDA (Python/Jupyter or Excel)

- Descriptive stats per branch/category
- Correlation: median income vs. revenue, population vs. revenue
- Per-capita and per-healthcare-center normalized metrics (create these as calculated fields — this is the client's explicit ask)
- Time series trend + basic seasonality check
- Outlier/data quality notes

3. Power BI Dashboard — suggested pages

- **Overview:** Total revenue, total boxes, avg order value, YTD trend, branch map/table
- **Branch Performance:** Revenue, revenue per capita, revenue per healthcare center, ranked
- **Category Analysis:** Category mix by branch, topical vs. non-topical split, top/bottom categories
- **Sales Rep Performance:** Rep $\times$ branch matrix, top performers

- **Payment & Demographics:** Payment method mix, avg order value by payment method, income vs. revenue scatter
- Global slicers: Branch, Category, Date range, Payment Method

4. **1-2 page written summary**

- Lead with the 3-4 most decision-relevant findings (client cares about *decisions*, not chart count)

Use this as your working brief. If you want, I can help you build out the Power Query steps, the DAX measures (per-capita revenue, MoM %, etc.), or review your dashboard layout against this brief once you have a draft.